

SHEIK MD AL ARIFIN

INDUSTRIAL AND PRODUCTION ENGINEER

Purbo Shoshaliya, Doshghoriya-3878, Chatkhil, Noakhali

01749427160 sk.alarifin@gmail.com Sheik Md Al Arifin Portfolio

PROFILE SUMMARY

Final-year Industrial & Production Engineering student interested in Operations, Supply Chain, and Data Analytics. Skilled in Python, Excel, Power BI, simulation, and optimization, with a focus on data-driven problem solving, operational efficiency, and continuous improvement. Seeking an entry-level role to apply and develop these skills.

EDUCATION

Shahjalal University of Science and Technology Sylhet, Bangladesh

BSc. in Industrial and Production Engineering | Current CGPA: 3.76/4.00 Jan 2023 – Present

Relevant coursework: Manufacturing Processes, Advanced Manufacturing System, Engineering Statistics, Industrial Management, Project Management, Operations Management, Operation Research, Computer Aided Drawing and Drafting, Product Design and Development, Quality Control and Management, Facility Layout and Design.

Chattogram Government Model School & College Chattogram, Bangladesh

Higher Secondary School in Science | GPA: 5.00/5.00 Jul 2018 – Jan 2021

WORK EXPERIENCE

Tech er Haat Bangladesh

Co-Founder & Business Analyst Jan 2025 – Present

- Managed sales accounts, product pricing, and inventory records for for 30+ electronic products, tracking stock levels and sales data to support purchasing and business decisions.

Bangladesh Industrial Technical Assistance Center (BITAC) Chattogram, Bangladesh

Industrial Training Duration: 12 days

- Completed 12-day industrial training covering industrial engineering and production control, with practical exposure to conventional manufacturing processes, production planning, and industrial operations.

PROJECTS

Simulation of a Sewing Line

Discrete-Event Simulation | Arena Rockwell

- Modeled an 18-process RMG line from time-study data on 26 workers; 150/150 throughput over 10 replications, 617.8 sec avg. cycle time, 3 bottlenecks identified.

Design and fabrication of a spur gear reversing mechanism

CAD Design & FEA Simulation | SolidWorks

- Designed and FEA-validated a lever-actuated gear reversing mechanism, achieving factors of safety of 1.26 (forward) and 6.22 (reverse), then fabricated a working 3D-printed prototype.

SKILLS

- **Simulation & Design:** Arena Rockwell, AutoCAD, COMSOL Multiphysics
- **Programming & Analytics:** Python, Power BI, Tableau, MS Excel, Power Query, MATLAB, Minitab
- **General Software:** MS Word, MS PowerPoint, Canva
- **Operations & Improvement:** Line Balancing, Productivity Improvement, Lean Manufacturing

RESEARCH PAPERS

- **Conference Paper (2025)** – "A Bi-Objective Optimization Model for Real-World Task Allocation in RMG Sewing Lines: Capacity Maximization and Minimization", IEOM 2025 Bangladesh: 8th Industrial Engineering and Operations Management Conference, World University of Bangladesh (WUB), Bangladesh.
- **Conference Paper (2025)** – "Data-Driven Prediction and Recommendation of Elective Courses for Master's Students", IEOM 2025 Bangladesh: 8th Industrial Engineering and Operations Management Conference, World University of Bangladesh (WUB), Bangladesh.

ORGANIZATIONAL EXPERIENCE

IPE Association, SUST Sylhet, Bangladesh
IT Secretary Sep 2026 – Present
IPE Association is a student organization at SUST dedicated to promoting academic and professional growth in the field of IPE.

- Coordinated IT, procurement, and event operations for the departmental farewell and managed technical activities for a SolidWorks training session.

IPESphere Season 2 Sylhet, Bangladesh
Design team Coordinator Aug 2026 – Sep 2026
The biggest national competition arranged by SUST IPE

- Coordinated a 10+ member team and personally delivered 20+ event design assets, managing task allocation, quality review, revisions, and final delivery.
- Led the CAD Competition Team during the Grand Finale, coordinating team members and ensuring smooth on-ground execution of competition activities.

CERTIFICATIONS

- Data Analyst — Datacamp
- Python for Data Science, AI & Development — IBM via Coursera
- Six Sigma Yellow Belt — Council for Six Sigma Certifications (CSSC)
- Power BI Fundamentals — Datacamp
- CSCMP Supply Chain Foundations — CSCMP via LinkedIn Learning
- Everyday Excel, Part 1 — University of Colorado Boulder via Coursera

ACHIEVEMENTS

- Finalist in **IPE Lean Sprint 3.0**, Bangladesh University of Engineering and Technology (BUET)
- Finalist in **IPE CaseQuest 2.0**, Inter Departmental Case Competition
- Selected among the top 12 teams in **UIHP SUST IC6** and completed a three-month training program.